

Lab 06

Exercises

1. Write an assembly language program that continuously reads a sequence of characters from user until the ENTER key pressed, and then counts and prints only the number of operators (+, -, /, *) within that string/sequence.

```
Enter a sequence of character: Abc+-Xyz/*  
Number of operator: 4
```

2. Write an assembly language program that continuously reads a sequence of characters from user until the ENTER key pressed, and then counts and prints only the capital/uppercase alphabets/characters.

```
Enter a sequence of characters: I have Entered Some Characters  
Number of Capital characters: 4
```

3. Write an assembly language program that continuously reads a sequence of characters from user until the ENTER key pressed, and then counts and prints only the small/lowercase alphabets/characters.

```
Enter a sequence of characters: Some Text  
Number of Small characters: 6
```

4. Write an assembly language program that continuously reads a sequence of characters from user until the ENTER key pressed, and then separately counts and prints the small/lowercase as well as capital/uppercase alphabets/characters.

```
Enter a sequence of characters: SomeText  
Number of Small characters: 6  
Number of Capital characters: 2
```

5. Write an assembly language program that asks the user to input two numbers and an operator (+ or -). After that it should display the result as per the entered numbers and the chosen operator, as shown below:

Case 1

```
Enter a number: 5
Enter a number: 3
Enter the operator (+ or -): -
Subtraction is: 2
```

Case 2

```
Enter a number: 4
Enter a number: 3
Enter the operator (+ or -): +
Addition is: 7
```

6. Repeat the above-mentioned program and show this menu in a loop and ask the user if he/she wants to repeat that operation. If the user enters **Y** or **y** (uppercase or lowercase), then the program should repeat until the user enters some other character (other than **Y** or **y**).
7. Write an assembly language program that continuously reads a sequence of characters from user until the ENTER key pressed, and then prints it in the reverse order.

Hint: Use the stack.

```
Enter a text string and press Enter: I am learning Assembly
The entered string in reverse order is: ylbmessA gnirael ma I
```

8. Write an assembly language program that continuously reads a sequence of characters from user until the ENTER key pressed, and stores those characters in Stack. After that, it should count and print the integer numbers only (numbers from 0 to 9).

```
Enter a sequence of character: Abc123Xyz098
No. Of Integers: 6
```